D3a – Annex O-Plus (Revised Version 2.0)

Operational Deep Layer: Approximation, Limitation, and Reversible Steering under Real Complexity

1. Status and Function

D3a belongs to the operational layer of the set.

It supplements D3 with the operational deep layer required to make the logic formulated there effective, viable, and institutionally implementable under real conditions.

It contains:

- no new axioms,
- no new normative setting,
- no alternative critique system.

Its function is:

to organize the operational approximation to the admissibility conditions defined in D2 in such a way that it remains stable under real complexity, limited examinability, and institutional time pressure.

2. Starting Point

D2 formulates three non-compensable admissibility questions:

- Path Integrity
- Irreversibility
- Externalization

D3 shows:

These quantities are not directly measurable.

It follows from this:

The operational task does not consist in establishing certainty, but in acting under uncertainty in such a way that structural maldevelopment becomes visible early and correction remains possible.

3. Operational Basic Problem

Real systems operate under:

- time pressure
- economic pressure
- political pressure
- incomplete information
- high complexity
- growing system dynamics
- increasing self-transformation

This creates a field of tension:

D2 demands structural admissibility

Reality forces decisions under uncertainty

4. Operational Basic Principles

4.1 Approximation instead of Completeness

No system can fully map the structural conditions.

The goal is a sufficiently robust approximation.

4.2 Limitation before Optimization

Optimization is subordinate.
Primary is:
Avoidance of structurally inadmissible development.

4.3 Effectiveness before Perfection

An incomplete but effective system is preferable to a perfect but practically unusable system.

4.4 Prioritization before Overloading

Only what contributes to securing the structural conditions is made mandatory.

4.5 Reversible Learning Loops

Development occurs in limited, recoverable steps.

4.6 HOLD as Central Steering Principle

In cases of unclear admissibility, optimization does not occur; instead, the process is stopped, limited, or transferred to controlled clarification.

5. The HOLD Mechanism (Core Element)

5.1 Definition

HOLD is the operational state
in which no irreversible further development occurs
because structural admissibility has not been sufficiently clarified.

5.2 Triggers for HOLD

HOLD is triggered when:

- indications of path narrowing appear
- indications of irreversible binding become visible
- externalization cannot be excluded
- central metrics are contradictory
- dissent exists between examination instances
- drift cannot be excluded
- human connectability is endangered

5.3 Meaning of HOLD

HOLD is:

- not a failure,
- not a standstill,
- not a refusal of progress.

HOLD is:
a transitional state for clarification under limitation.

5.4 Operational Consequences of HOLD

In the HOLD state the following applies:

- no scaling
- no irreversible integration
- no expansion of effect
- no further condensation of critical structures

Instead:

- targeted additional examination
- expansion of the evidence base
- variation of examination conditions

- reduction of the scope of intervention
- return to smaller, controllable units

6. Operational Decision Space

D3a distinguishes three operational states:

6.1 GO (Approval)

Prerequisite:

- no sufficient indications of structural violation
- sufficiently stable evidence

Measure:

- limited further development
- controlled scaling

6.2 HOLD (Clarification)

Prerequisite:

- uncertainty or contradictory evidence

Measure:

- no irreversible decision
- additional examination

6.3 STOP (Block)

Prerequisite:

- structural violation probable

Measure:

- no further pursuit
- if necessary, rollback or isolation

7. Operational Core Variables

For condensing the examination, D3a uses a prioritized core set:

- **tc_rep –** Stability of the representation
- **tc_path –** Path fidelity
- **tc_total –** Aggregated consistency
- **drift –** Deviation between visible behavior and internal structure
- **path_risk / PD –** Risk of structural narrowing
- **Rev –** Reversibility
- **CI –** Integrity without Externalization
- **human_core_ok –** Preservation of human agency
- **dissent_risk –** Loss of contradiction
- **ws_struct –** Structural viability

Important:

These variables are not truth.

They are operational approximations.

Additionally, the following can be maintained as domain-specific auxiliary variables:

- **resonance_quality –** Quality of effective, selective, and context-stable coupling
- **boundary_integrity –** Preservation of functional inner-outer distinction with simultaneously mediated exchange
- **binding_profile –** Extent and quality of viable interdependency formation
- **redundancy_synergy_balance –** Ratio between robust safeguarding and higher-level

emergence
- **metastability_range –** Mobility between integration and autonomy without rigidity or decay
- **attention_integrity –** Preservation of human attention, reflection, and judgment capacity under system use

These variables are not new norms and not independent approval variables. They are operational approximations that may only be evaluated in connection with the D2 conditions and the O4/O6 gating.

8. Operational Steering Logic

8.1 Sequential Examination

1. Admissibility examination (Gate)
2. Multidimensional classification
3. Limited validation

8.2 No Single-Metric Decision

No individual key figure may be decisive.

8.3 Escalation Logic

With increasing:

- effect
- scaling
- irreversibility

the following must increase:

- examination depth
- examination diversity
- examination strictness

9. Self-Transforming Systems (Extension)

Special requirements apply to systems that:

- improve themselves
- change their own structure
- adapt their own decision rules

9.1 Additional Risks

- covert drift
- examination anticipation
- decoupling from reality
- acceleration of maldevelopment

9.2 Operational Consequences

For such systems the following applies:

- narrower HOLD thresholds
- shorter iteration cycles
- stronger validation requirements
- mandatory multi-instance examination
- explicit drift monitoring

10. Relation to D3

D3 defines:

- Structure of the examination

D3a defines:

- Implementation of the examination

11. Relation to D4 / D4a

D3a presupposes:

- human connectability remains preserved
- decisions remain traceable
- agency is not silently replaced

12. Relation to D6

D3a operates under the assumption:

Every protective mechanism can fail.

Therefore the following applies:

- continuous counter-examination
- no ultimate certainty
- permanent revisability

13. Limits

D3a does not guarantee:

- error-freeness
- complete control
- stable predictability

D3a guarantees only:

- limitation of maldevelopment
- preservation of correctability
- visibility of structural risks

14. Closing Sentence

Operational steering in the solution space means:

not maximum speed,

not maximum efficiency,

not maximum performance,

but:

maximum preservation of structural admissibility under real conditions.